

Scalability & Future Plan

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to local deployment	Cannot handle large number of users	High
2	No real-time weather data integration	Predictions may lack real-time accuracy	Medium
3	Basic UI and limited user interaction	Affects user experience	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users (local system)	Deploy on cloud (AWS/Render) with load balancing
2	Data Storage	Uses static CSV dataset	Use cloud database (MongoDB / PostgreSQL)
3	Performance	Single ML model processing	Optimize model & use caching for faster predictions
4	Security	Basic input validation	Add authentication & secure API handling

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Weather API Integration	0–1 months	More accurate real-time predictions
Phase 3	Mobile App Development	0–1 months	Increased accessibility for farmers
Phase 4	Fertilizer Recommendation System	1–2 months	Complete smart agriculture solution